PES UNIVERSITY

Department of Computer Science & Engineering

Digital Forensics and Incident Responses

UE23CS342BA6

Assignment 3A Submission

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Topic: Unit 1 - Lab 3A: Deleted Data RECOVERY

UE23CS342BA6

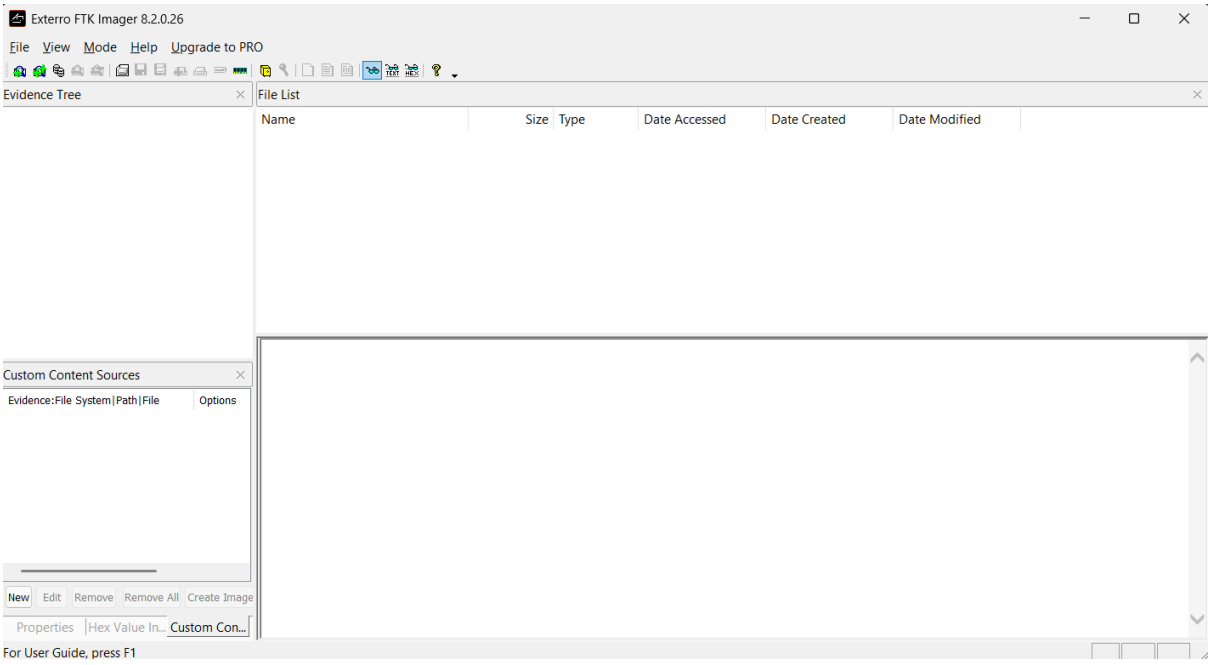
Learning Objective: To demonstrate the ability to recover deleted data using data recovery tools and manually carving for files based on header and footer information from unallocated space.

Narrative: Recovering deleted data is one of the tasks we are most often asked to perform as digital forensic analysts. It is also a great party trick. In this exercise we will be using tools and manual processes to recover deleted data.

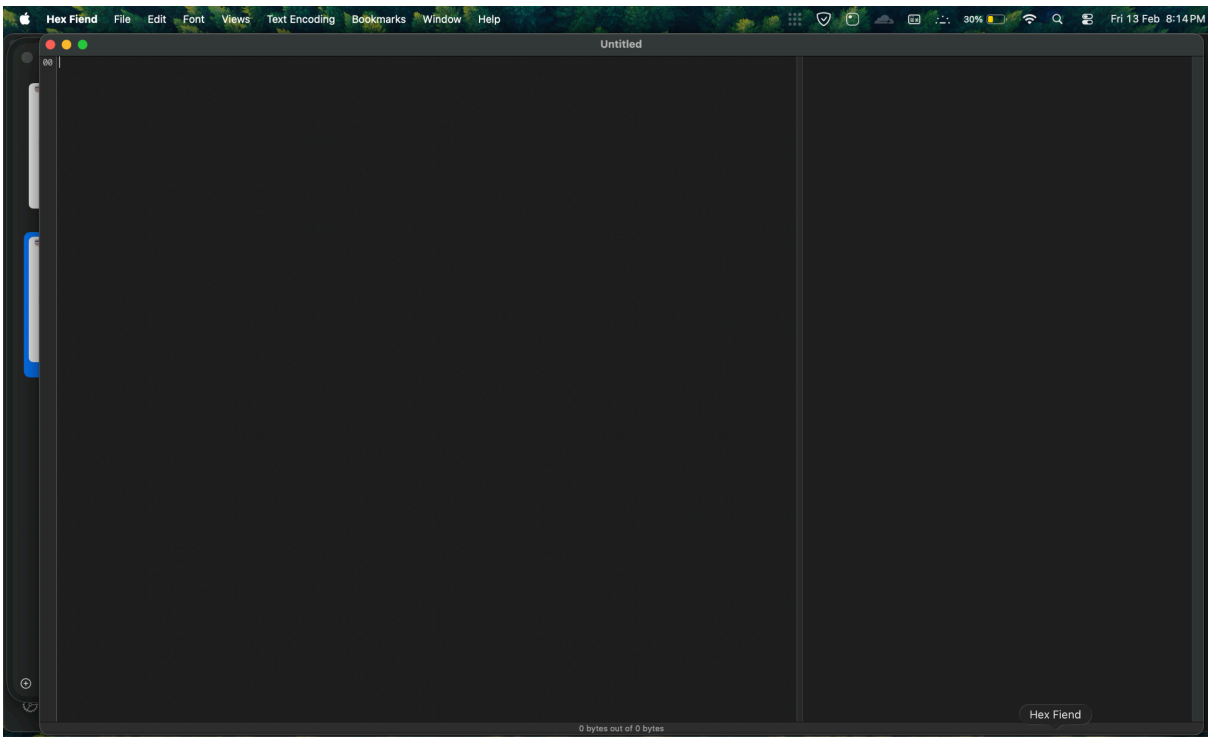
Part A: Setup

1. FTK Imager

- Download and install FTK from the official site.
- Launch FTK once installation completes.



2. Download Recuva
3. Download HxD (Hex editor)/ HexFiend



4. Download PhotoRec (from testdisk bundle)m

Part B: Create a working folder on your desktop:

1) Lab3/Lab 3A/
2) USB1/
3) USB2/
4) USB3/
5) USB4/

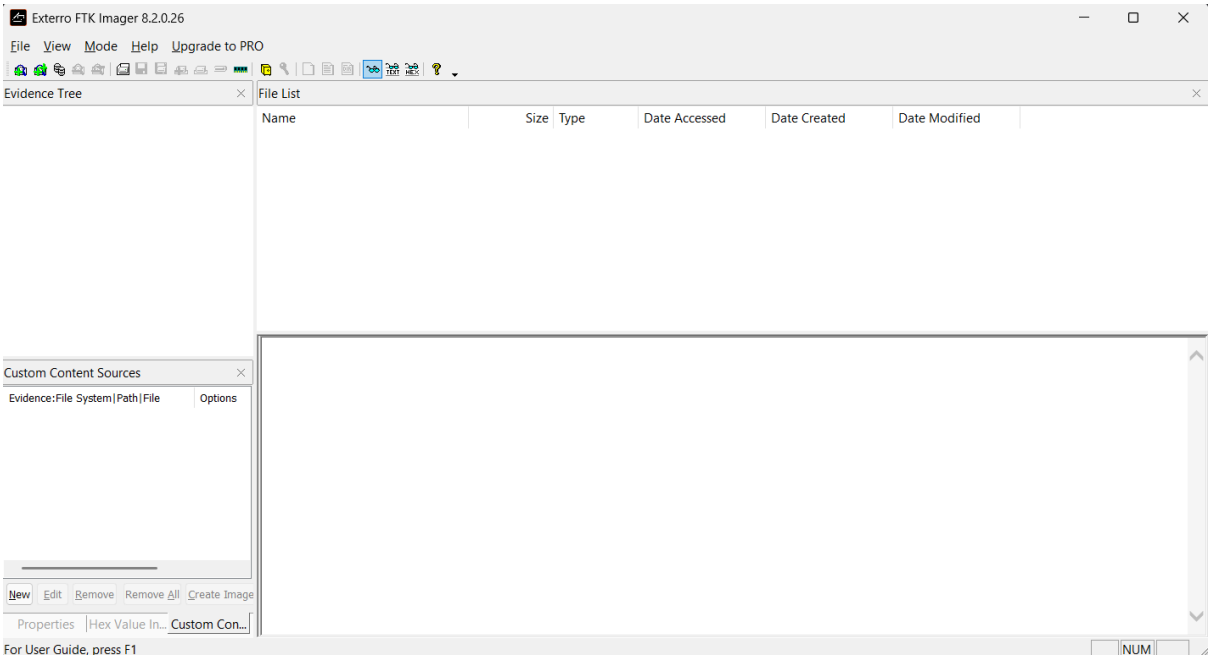
Part C: Inspect the USB files and perform the tasks

USB1.E01
Narrative: A suspect has been accused of traveling to Dubai for arms deals. The subject denies any involvement in arms, or any travel/interest in Dubai. A thumb drive was seized from the subject, which appears to be blank. A forensic image of the thumb drive has been created (USB1.E01), and you are asked to recover any deleted data from the image.

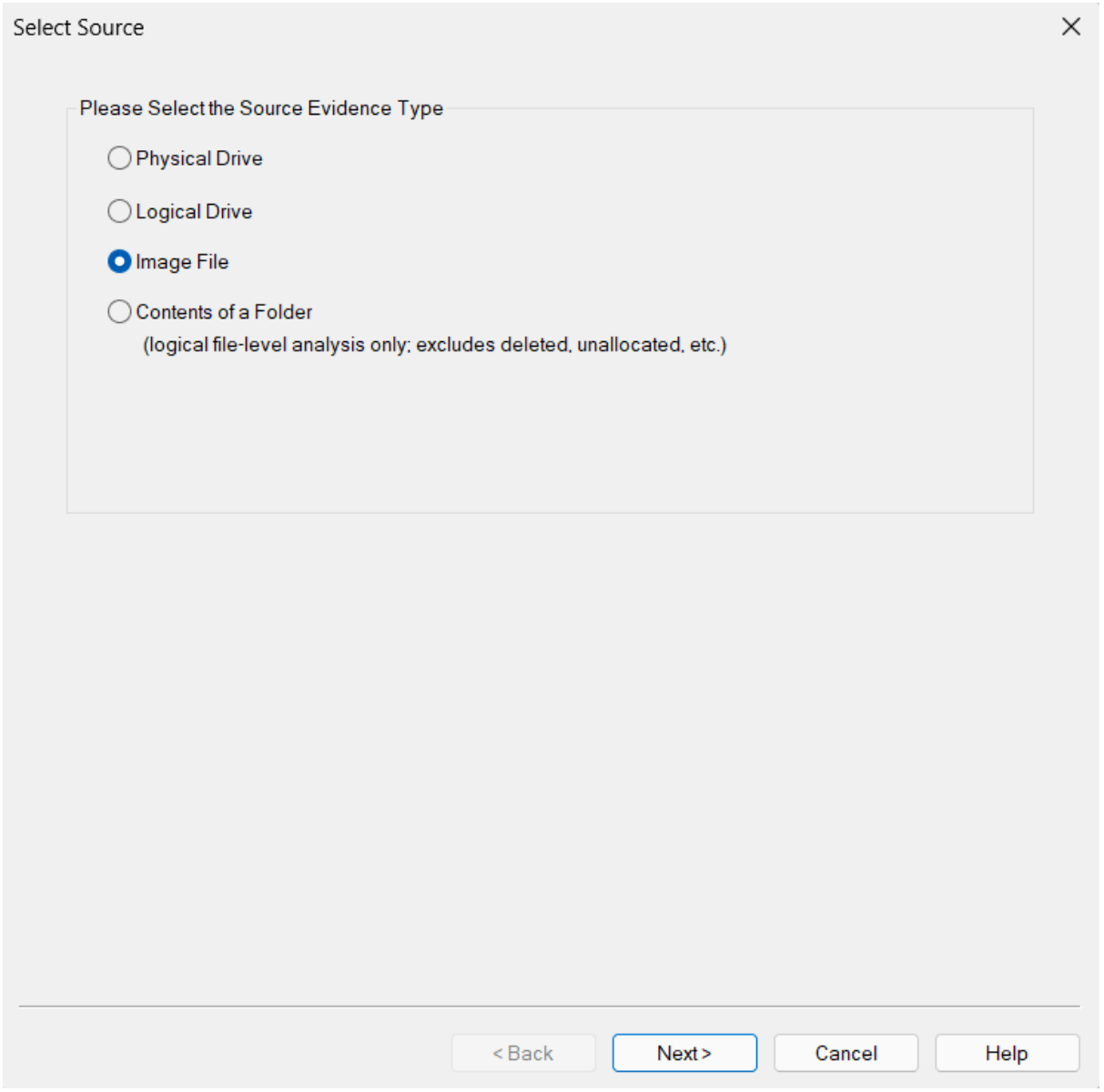
1) Mount Image

Step1: Open FTK Imager

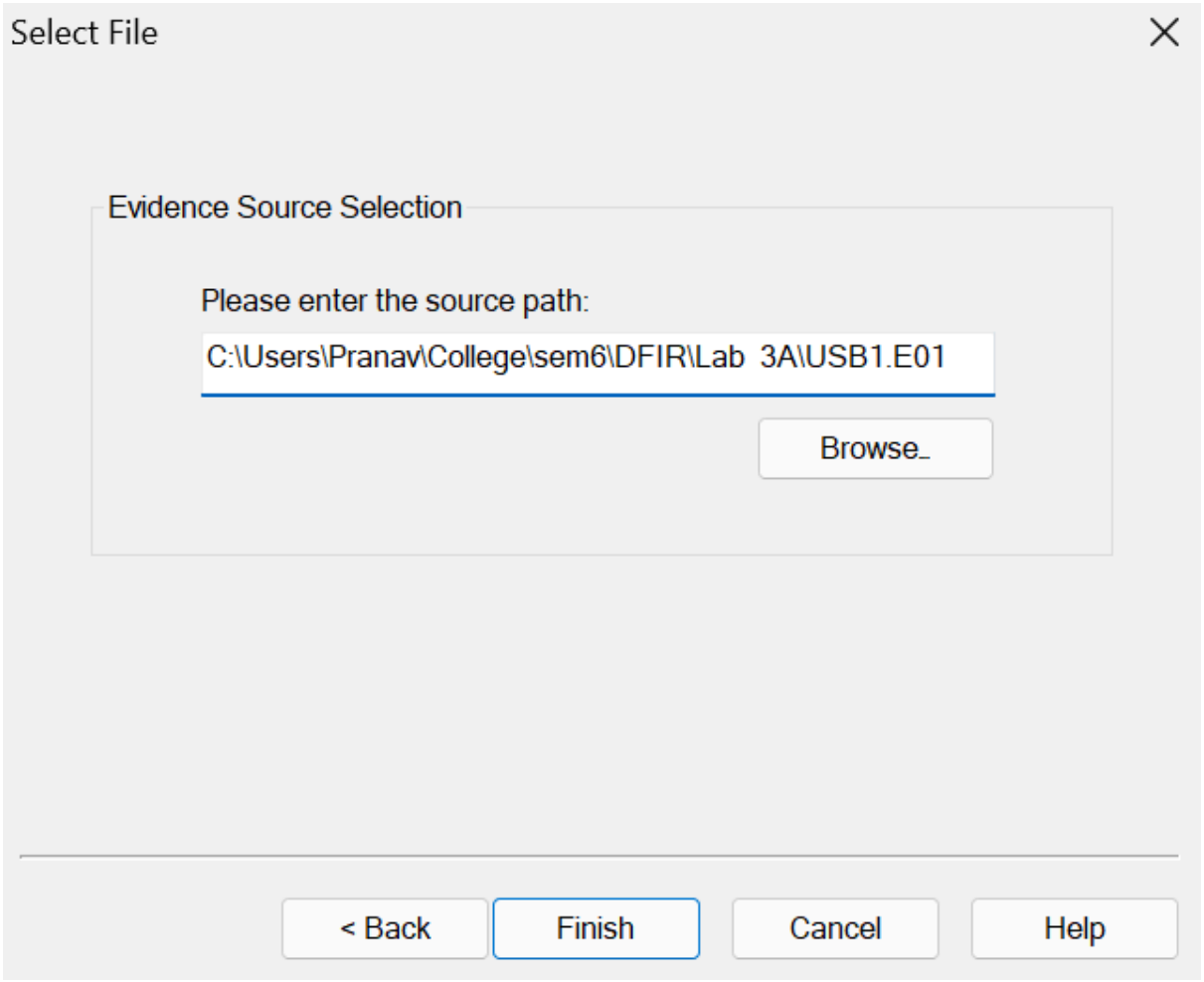
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Step2: Select Source Type = Image File

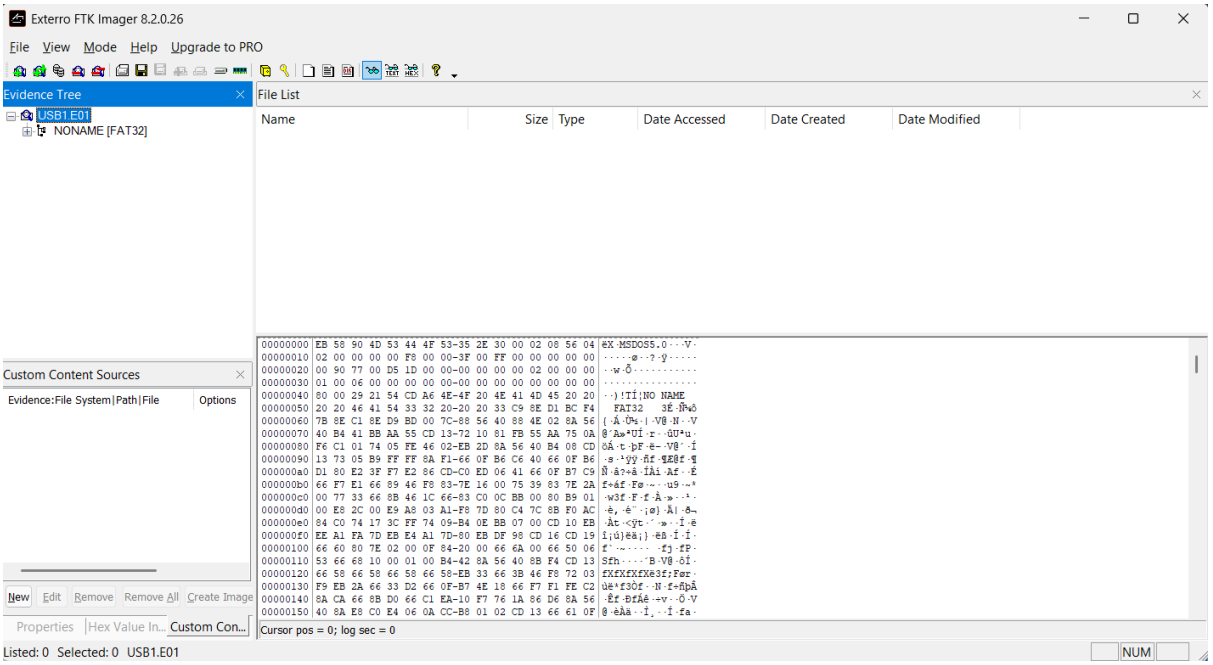


Step3: Select Image File Source = USB1.E01

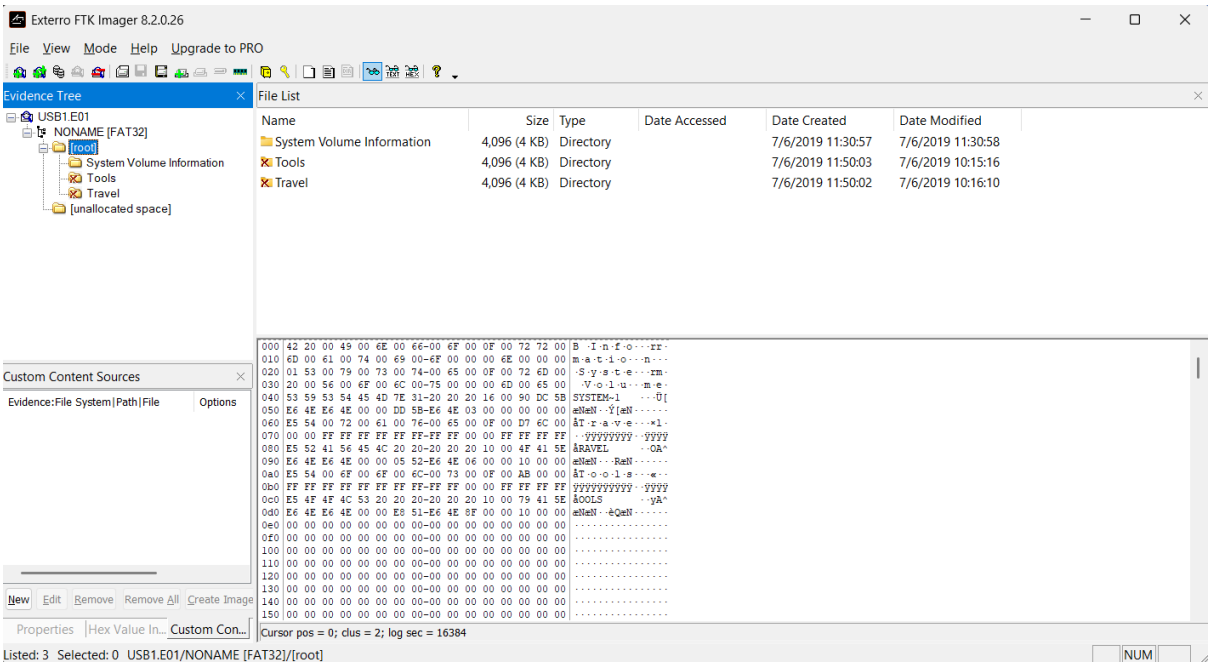


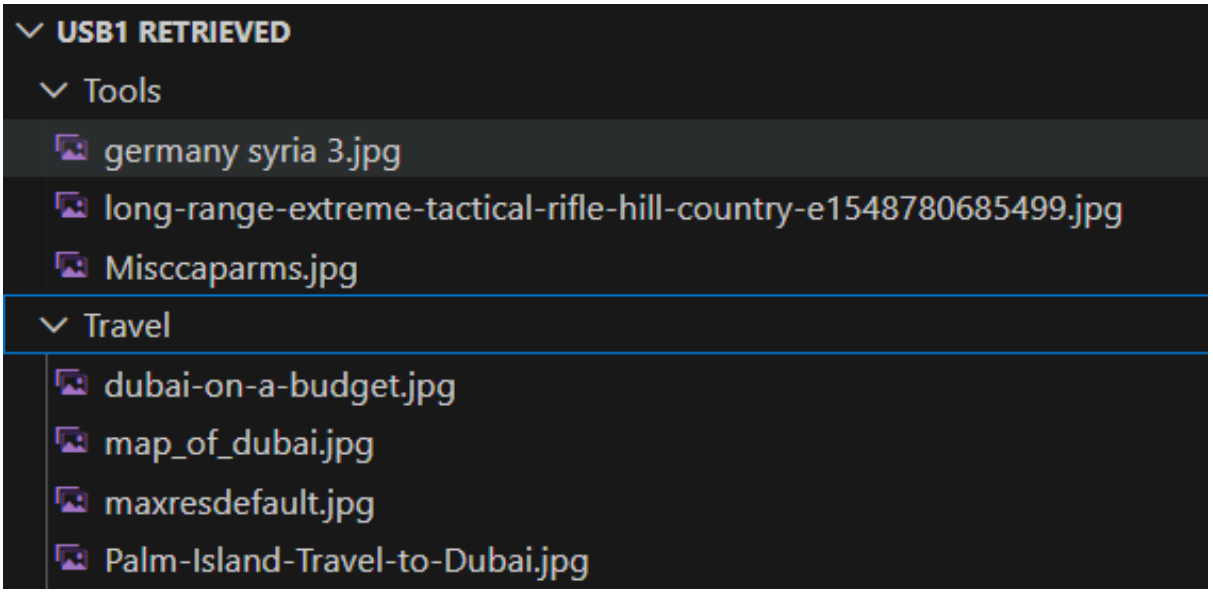
Tasks:

1. What type of file system is on this drive?
   Answer: The file system here is FAT32



2. Using the program FTK Imager, examine this image for previously deleted images.
   Answer: Directories with the X contain the deleted images.





3. List any deleted files of value found on this device – include file name and parent directory.
   Answer: List of deleted files are:
   Tools/:
   A. germany syria 3.jpg
   B. long-range-extreme-tactile-rifle-hill-country-e1548780685499.jpg
   C. Misccaparms.jpg

   Travel/:

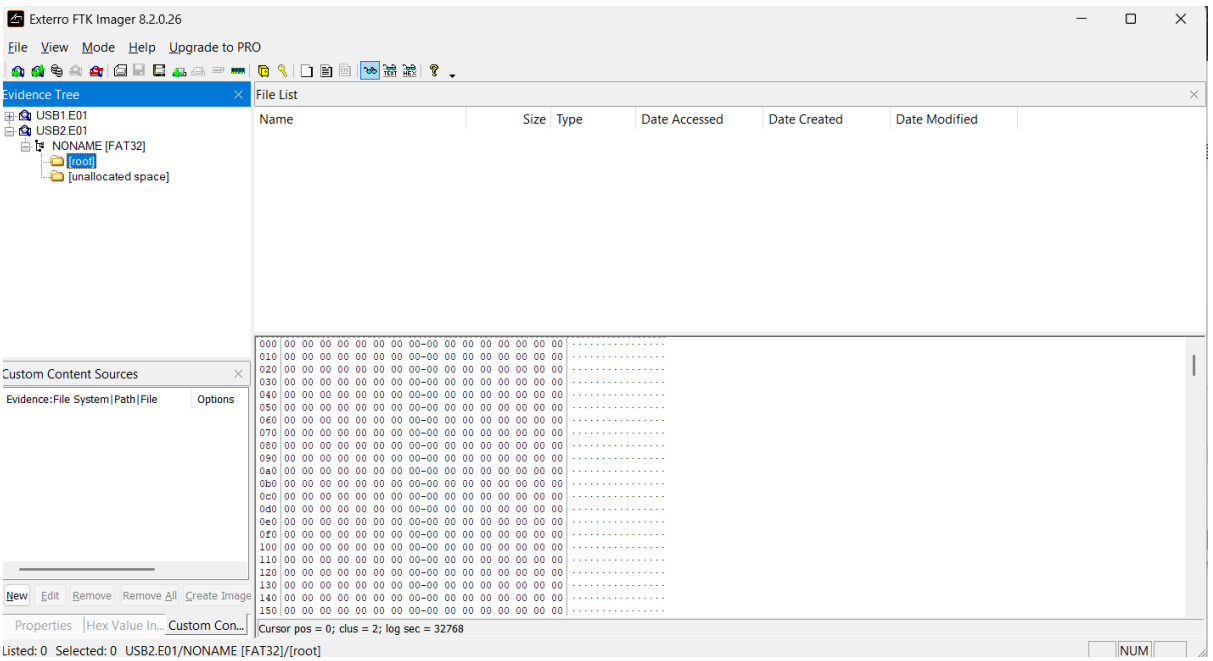
   A. dubai-on-a-budget.jpg
   B. Map_of_dubai.jpg
   C. maxresdefault.jpg
   D. Palm-Island-Travel-to-Dubai.jpg

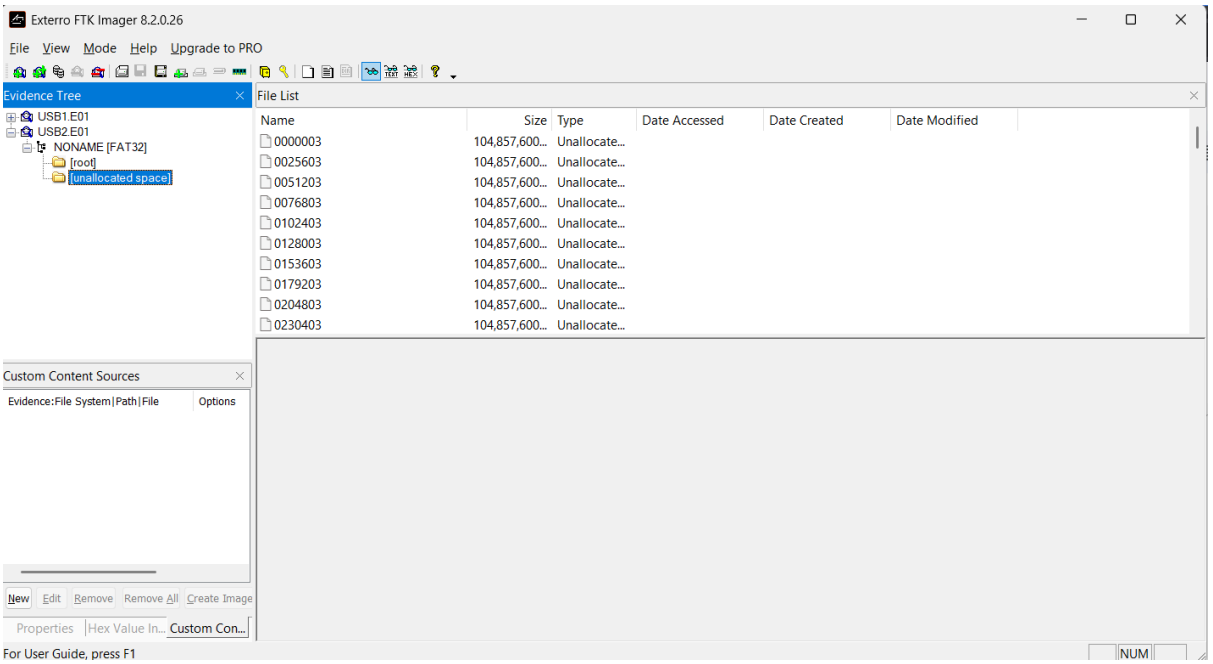
USB2.E01

Narrative: A suspect has been accused of trying to obtain microbrewery equipment to manufacture Anthrax. The subject was found to be in possession of a thumb drive, which appears to be blank. A forensic image of the thumb drive has been created (USB2.E01), and you are asked to recover any deleted data from the image.

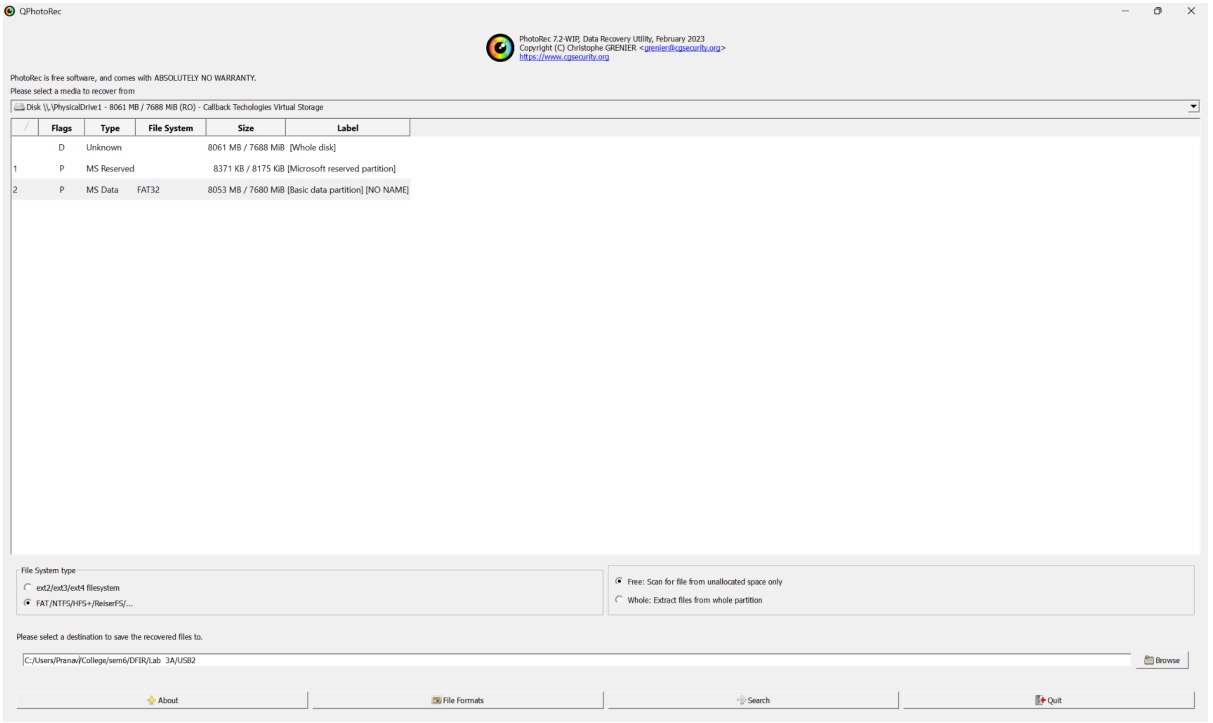
Tasks:

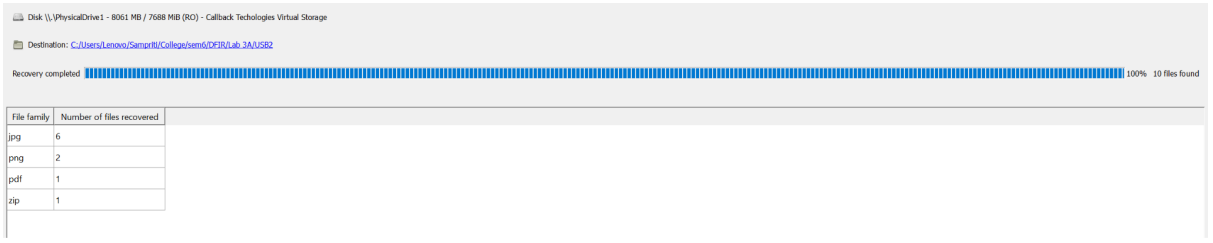
1. What do you see using FTK Imager?
   Answer: The root directory contains no files. The majority of the storage appears to consist of unallocated space. The drive is formatted using the FAT32 file system.

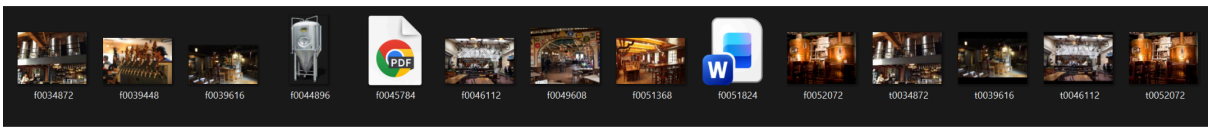




2. Using the program PhotoRec, examine this image for previously deleted files.
   Answer: An in-depth scan of the unallocated space was conducted using PhotoRec, resulting in the successful recovery of deleted files through file carving.The screenshots below show the steps performed along with the deleted files recovered.







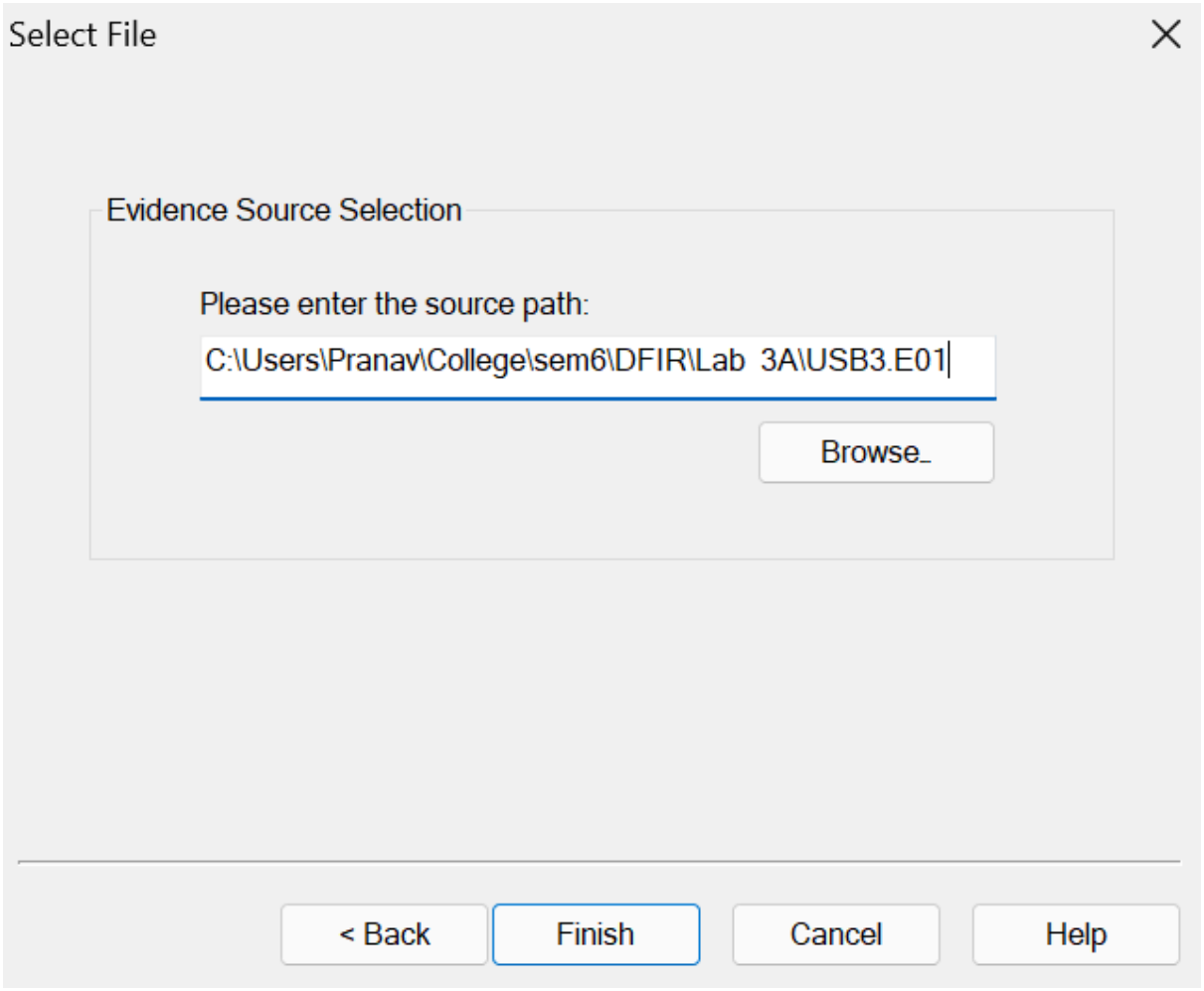
3. Zip any recovered files and submit along with this document as USB2.zip.

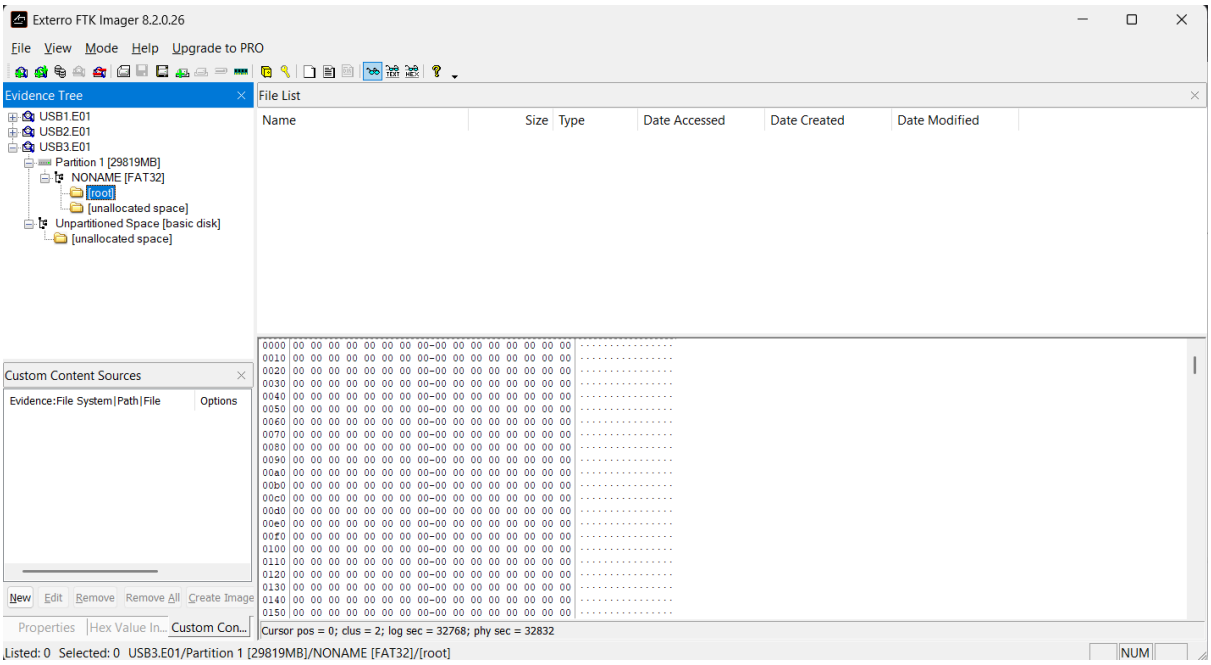
USB3.E01

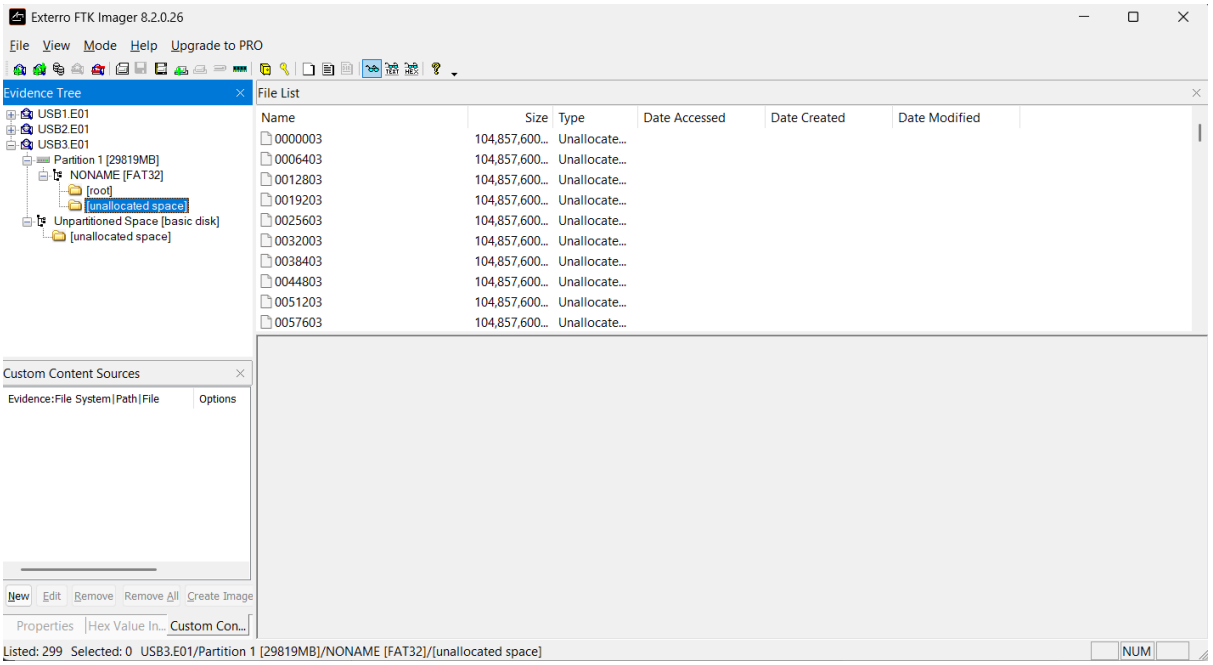
Narrative: A suspect has been accused of smuggling several shipments of cocaine using small human powered boats. The subject was found to be in possession of a thumb drive, which appears to be blank. A forensic image of the thumb drive has been created (USB3.E01), and you are asked to recover any deleted data from the image.

Tasks:

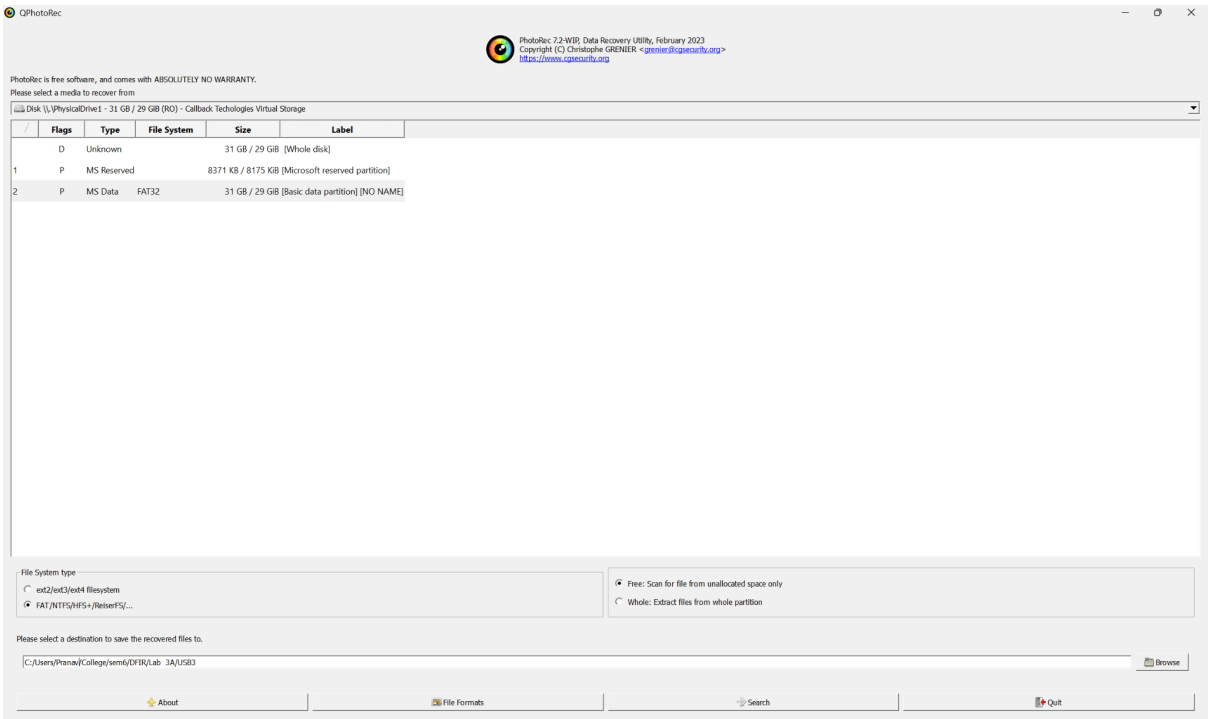
1. What do you see using FTK Imager?
   Answer:

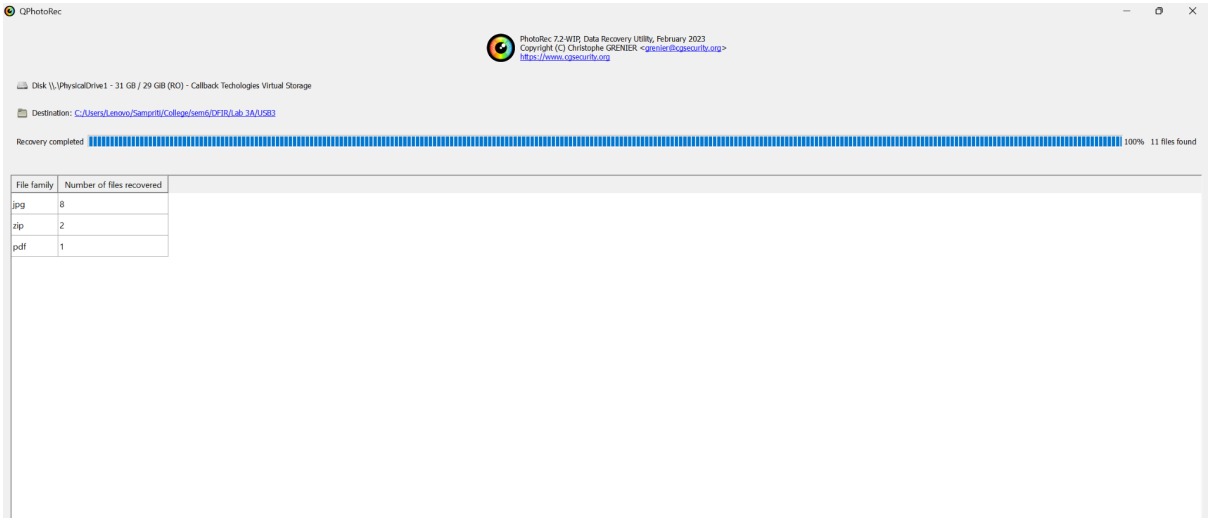

Select File
Evidence Source Selection
Please enter the source path:
C:\Users\Pranav\College\sem6\DFIR\Lab 3A\USB3.E01
Browse_
< Back
Finish
Cancel
Help

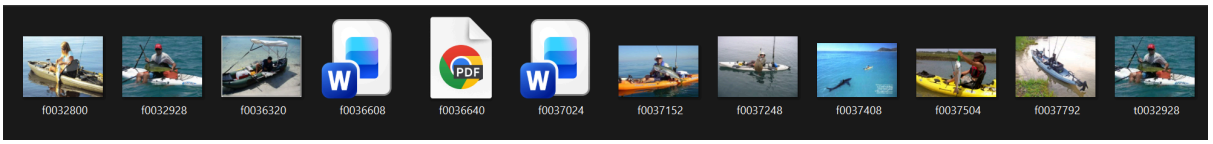

Exterro FTK Imager 8.2.0.26
File View Mode Help Upgrade to PRO
Evidence Tree
USB1.E01
USB2.E01
USB3.E01
Partition 1 [29819MB]
NONAME [FAT32]
[root]
[unallocated space]
Unpartitioned Space [basic disk]
[unallocated space]
File List
Name
Size
Type
Date Accessed
Date Created
Date Modified
Custom Content Sources
Evidence:File System|Path|File
Options
New
Edit
Remove
Remove All
Create Image
Properties
Hex Value In...
Custom Con...
Cursor pos = 0; clus = 2; log sec = 32768; phy sec = 32832
Listed: 0 Selected: 0 USB3.E01/Partition 1 [29819MB]/NONAME [FAT32]/[root]
NUM



2. Using the program PhotoRec, examine this image for previously deleted files.
   Answer: PhotoRec conducted a comprehensive scan of the unallocated space and successfully retrieved several deleted files. The steps followed and the recovered files can be seen in the below screenshots.







3. Zip any recovered files and submit along with this document as USB3.zip.

USB4.E01

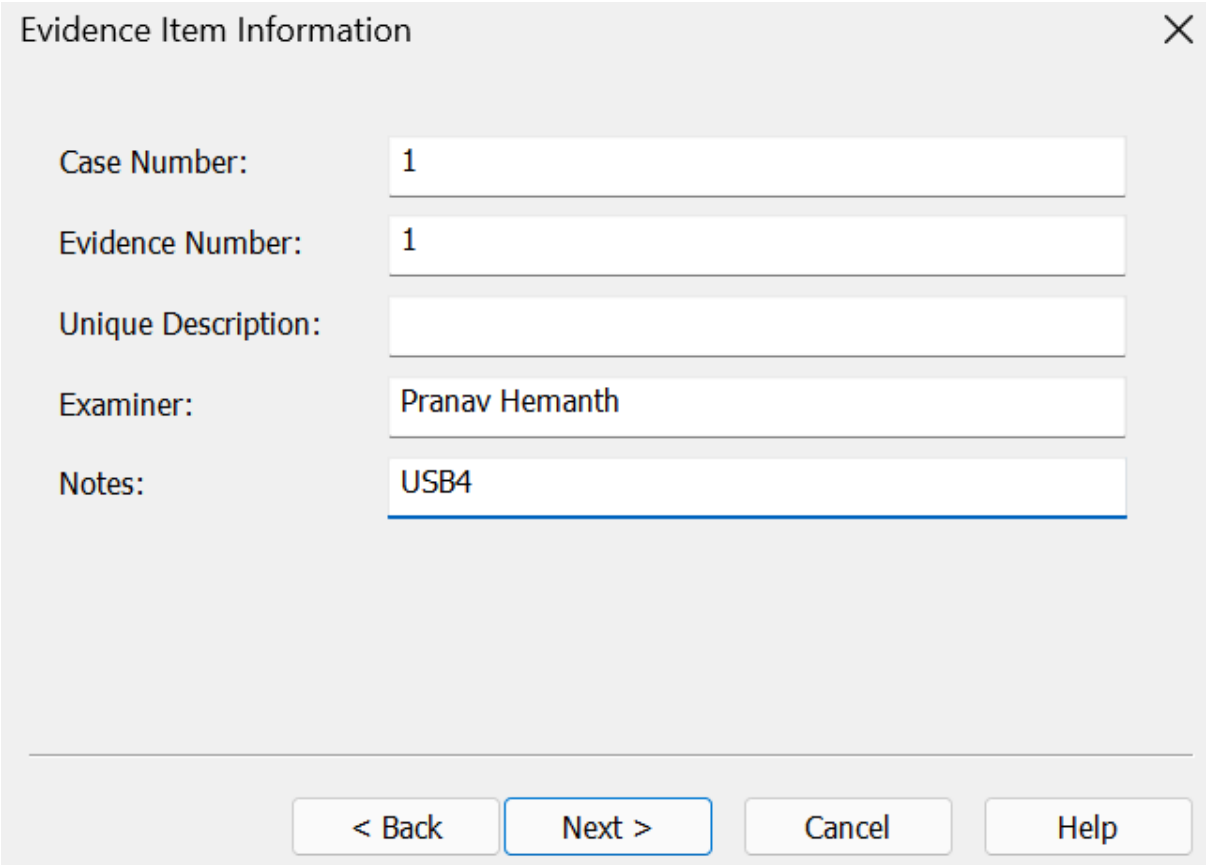
Narrative: This usb was found in a secure facility. It is believed that a document existed on this device at some point that is of value. It is believed that the file type is a PDF file. Research the file type and discover the file header on the image using a hex editor (like HxD).

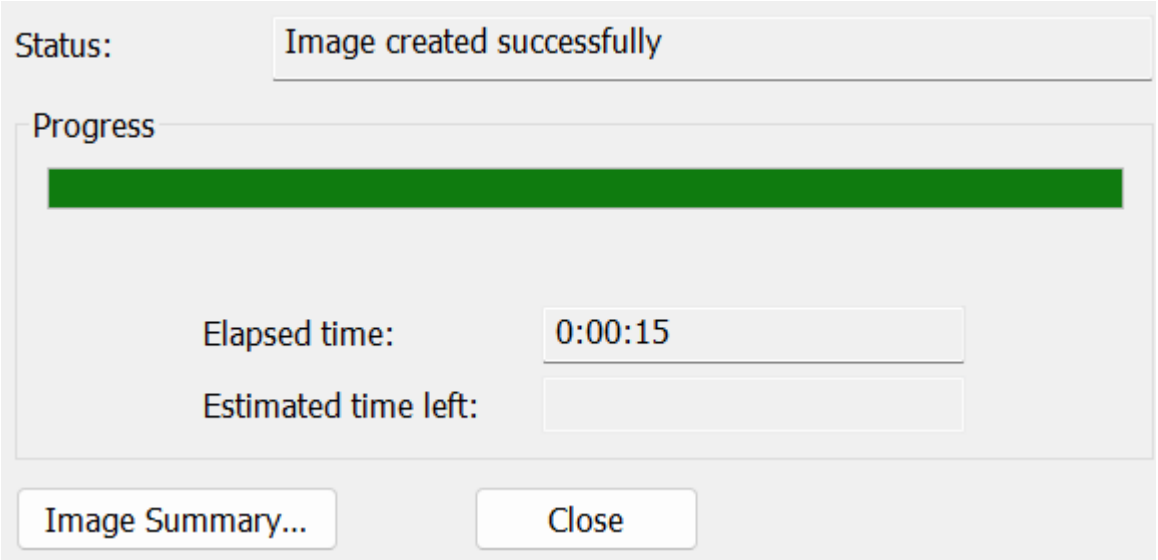
Tasks:

1. Do not use the automated tools, recover this file using nothing more than HxD.
2. Give a brief description of the contents of this file.
   Answer: The recovered PDF appears to describe a purported “classified facility” and an objective to penetrate it. It contains narrative instructions directing the reader to obtain restricted information, accompanied by images of what is in reality a Jelly Belly production plant. The document includes exterior photographs of the site, mentions of security personnel in a humorous tone, and interior images of the manufacturing area where jelly beans are processed and sorted. Overall, the file resembles a satirical classified mission brief focused on the Jelly Belly factory and its operations.
3. Submit this file along with this document as USB4.PDF

To do this we will first export raw image from FTK following the below steps:

1. Open FTK Imager
2. Add USB4.E01
3. Right-click on USB4.E01
4. Select Export Disk Image
5. Choose Image Type = Raw (dd)
6. Save as USB4.dd

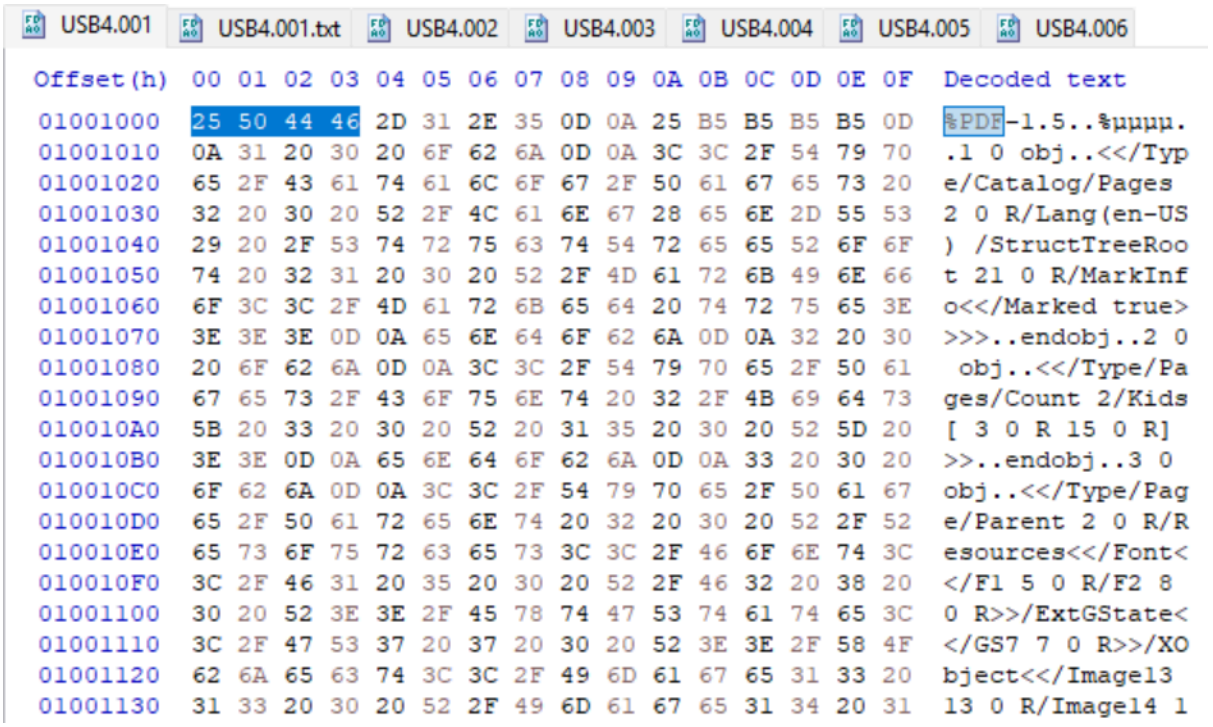




Next we will open this raw image in HxD and follow the below steps:

Open Raw Image in HxD

1. Open HxD
2. Select USB4.dd
3. Search for 25 50 44 46



USB4.001 | USB4.001.txt | USB4.002 | USB4.003 | USB4.004 | USB4.005 | USB4.006

Offset(h)	00	01	02	03	04	05	06	07	08	09	0A	0B	0C	0D	0E	0F	Decoded text
01001000	25	50	44	46	2D	31	2E	35	0D	0A	25	B5	B5	B5	B5	0D	%PDF-1.5..%µµµµ.
01001010	0A	31	20	30	20	6F	62	6A	0D	0A	3C	3C	2F	54	79	70	.1 0 obj..<</Typ
01001020	65	2F	43	61	74	61	6C	6F	67	2F	50	61	67	65	73	20	e/Catalog/Pages
01001030	32	20	30	20	52	2F	4C	61	6E	67	28	65	6E	2D	55	53	2 0 R/Lang(en-US
01001040	29	20	2F	53	74	72	75	63	74	54	72	65	65	52	6F	6F	) /StructTreeRoo
01001050	74	20	32	31	20	30	20	52	2F	4D	61	72	6B	49	6E	66	t 21 0 R/MarkInf
01001060	6F	3C	3C	2F	4D	61	72	6B	65	64	20	74	72	75	65	3E	o<</Marked true>
01001070	3E	3E	3E	0D	0A	65	6E	64	6F	62	6A	0D	0A	32	20	30	>>>..endobj..2 0
01001080	20	6F	62	6A	0D	0A	3C	3C	2F	54	79	70	65	2F	50	61	obj..<</Type/Pa
01001090	67	65	73	2F	43	6F	75	6E	74	20	32	2F	4B	69	64	73	ges/Count 2/Kids
010010A0	5B	20	33	20	30	20	52	20	31	35	20	30	20	52	5D	20	[3 0 R 15 0 R]
010010B0	3E	3E	0D	0A	65	6E	64	6F	62	6A	0D	0A	33	20	30	20	>>..endobj..3 0
010010C0	6F	62	6A	0D	0A	3C	3C	2F	54	79	70	65	2F	50	61	67	obj..<</Type/Pag
010010D0	65	2F	50	61	72	65	6E	74	20	32	20	30	20	52	2F	52	e/Parent 2 0 R/R
010010E0	65	73	6F	75	72	63	65	73	3C	3C	2F	46	6F	6E	74	3C	esources<</Font<
010010F0	3C	2F	46	31	20	35	20	30	20	52	2F	46	32	20	38	20	</F1 5 0 R/F2 8
01001100	30	20	52	3E	3E	2F	45	78	74	47	53	74	61	74	65	3C	0 R>>/ExtGState<
01001110	3C	2F	47	53	37	20	37	20	30	20	52	3E	3E	2F	58	4F	</GS7 7 0 R>>/XO
01001120	62	6A	65	63	74	3C	3C	2F	49	6D	61	67	65	31	33	20	bject<</Image13
01001130	31	33	20	30	20	52	2F	49	6D	61	67	65	31	34	20	31	13 0 R/Image14 1

Next we carve the PDF by selecting the starting offset containing 25 50 44 46 (%PDF) to 25 45 4F 46 (%%EOF) and pasting it into a new file in HxD

```
Offset(h)  00 01 02 03 04 05 06 07 08 09 0A 0B 0C 0D 0E 0F  Decoded text
00000000   25 50 44 46 2D 31 2E 35 0D 0A 25 B5 B5 B5 B5 0D  %PDF-1.5..%µµµµ.
00000010   0A 31 20 30 20 6F 62 6A 0D 0A 3C 3C 2F 54 79 70  .1 0 obj..<</Typ
00000020   65 2F 43 61 74 61 6C 6F 67 2F 50 61 67 65 73 20  e/Catalog/Pages
00000030   32 20 30 20 52 2F 4C 61 6E 67 28 65 6E 2D 55 53  2 0 R/Lang(en-US
00000040   29 20 2F 53 74 72 75 63 74 54 72 65 65 52 6F 6F  ) /StructTreeRoo
00000050   74 20 32 31 20 30 20 52 2F 4D 61 72 6B 49 6E 66  t 21 0 R/MarkInf
00000060   6F 3C 3C 2F 4D 61 72 6B 65 64 20 74 72 75 65 3E  o<</Marked true>
00000070   3E 3E 3E 0D 0A 65 6E 64 6F 62 6A 0D 0A 32 20 30  >>>..endobj..2 0
00000080   20 6F 62 6A 0D 0A 3C 3C 2F 54 79 70 65 2F 50 61   obj..<</Type/Pa
00000090   67 65 73 2F 43 6F 75 6E 74 20 32 2F 4B 69 64 73  ges/Count 2/Kids
000000A0   5B 20 33 20 30 20 52 20 31 35 20 30 20 52 5D 20  [ 3 0 R 15 0 R]
000000B0   3E 3E 0D 0A 65 6E 64 6F 62 6A 0D 0A 33 20 30 20  >>..endobj..3 0
000000C0   6F 62 6A 0D 0A 3C 3C 2F 54 79 70 65 2F 50 61 67  obj..<</Type/Pag
000000D0   65 2F 50 61 72 65 6E 74 20 32 20 30 20 52 2F 52  e/Parent 2 0 R/R
000000E0   65 73 6F 75 72 63 65 73 3C 3C 2F 46 6F 6E 74 3C  esources<</Font<
000000F0   3C 2F 46 31 20 35 20 30 20 52 2F 46 32 20 38 20  </F1 5 0 R/F2 8
00000100   30 20 52 3E 3E 2F 45 78 74 47 53 74 61 74 65 3C  0 R>>/ExtGState<
00000110   3C 2F 47 53 37 20 37 20 30 20 52 3E 3E 2F 58 4F  </GS7 7 0 R>>/XO
00000120   62 6A 65 63 74 3C 3C 2F 49 6D 61 67 65 31 33 20  bject<</Image13
00000130   31 33 20 30 20 52 2F 49 6D 61 67 65 31 34 20 31  13 0 R/Image14 1
00000140   34 20 30 20 52 3E 3E 2F 50 72 6F 63 53 65 74 5B  4 0 R>>/ProcSet[
00000150   2F 50 44 46 2F 54 65 78 74 2F 49 6D 61 67 65 42  /PDF/Text/ImageB
00000160   2F 49 6D 61 67 65 43 2F 49 6D 61 67 65 49 5D 20  /ImageC/ImageI]
00000170   3E 3E 2F 4D 65 64 69 61 42 6F 78 5B 20 30 20 30  >>/MediaBox[ 0 0
00000180   20 36 31 32 20 37 39 32 5D 20 2F 43 6F 6E 74 65   612 792] /Conte
00000190   6E 74 73 20 34 20 30 20 52 2F 47 72 6F 75 70 3C  nts 4 0 R/Group<
000001A0   3C 2F 54 79 70 65 2F 47 72 6F 75 70 2F 53 2F 54  </Type/Group/S/T
000001B0   72 61 6E 73 70 61 72 65 6E 63 79 2F 43 53 2F 44  ransparency/CS/D
000001C0   65 76 69 63 65 52 47 42 3E 3E 2F 54 61 62 73 2F  eviceRGB>>/Tabs/
000001D0   53 2F 53 74 72 75 63 74 50 61 72 65 6E 74 73 20  S/StructParents
000001E0   30 3E 3E 0D 0A 65 6E 64 6F 62 6A 0D 0A 34 20 30  0>>..endobj..4 0
000001F0   20 6F 62 6A 0D 0A 3C 3C 2F 46 69 6C 74 65 72 2F   obj..<</Filter/
00000200   46 6C 61 74 65 44 65 63 6F 64 65 2F 4C 65 6E 67  FlateDecode/Leng
00000210   74 68 20 38 38 33 3E 3E 0D 0A 73 74 72 65 61 6D  th 883>>..stream
00000220   0D 0A 78 9C 8D 56 DB 6E DB 38 10 7D 37 E0 7F 98  ..xœ.VÛnÛ8.}7à.˜
00000230   47 71 11 33 22 45 51 D4 22 C8 43 AE 48 17 C5 6E  Gq.3"EQÔ"ÈC®H.Ån
00000240   51 03 C5 A2 D8 07 46 A6 6D 01 8E 83 4A 4A DD FC  Q.Å¢Ø.F¦m.ŽƒJJÝü
00000250   FD CE 8C 64 47 76 2D C3 49 E0 28 E4 5C CF 9C 33  ýÎŒdGv-ÃIà(ä\Ïœ3
00000260   0A 5C FE 03 57 57 97 9F 6F 9F EE 20 BE BE 86 9B  .\þ.WW—ŸoŸî ¾¾†›
00000270   BB 5B B8 99 8E 47 97 0F 0A 94 92 B1 81 E9 7C 3C  »[¸™ŽG—..”’±.é|<
00000280   52 10 E3 B7 82 4C CB 58 1B C8 E2 5C 5A 0B D3 17  R.ã·‚LËX.Èâ\Z.Ó.
00000290   B4 7B FC 9A C1 A2 1E 8F 62 58 D0 C7 E3 78 F4 3D  ´{üšÁ¢..bXÐÇãxô=
000002A0   9A 2E CB 1A F0 C7 83 C8 A3 62 E5 EB BA 9C 8B 24  š.Ë.ðÇƒȣbåëºœ‹$
```

Then we save this as USB4.pdf for submission